

Robust Quality Classification of Germinated Oil Palm Seeds under Real-World Distribution Shift

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Abstract. Self-supervised learning techniques were applied in this work to yield a ResNet50 encoder to be compared to a baseline 50-layer residual network (ResNet50) from supervised learning, on the downstream seed quality classification task across images with illumination and pose variations, without having been trained on those variations. Saliency maps are an interpretability tool for failure analysis of the baseline model. The most performant model in terms of accuracy, F1 and AUC was the online encoder trained on the BYOL loss function and fine-tuned on the downstream task. Adjusted Random Index (ARI) metric from K -means clusters ($K = 2$) of high dimensional embeddings showed that the cluster structure was better preserved across Batch-2 and 3 but was still showed degradation relative to Batch-1.

Keywords: Computer Vision · Environmental Shift · Augmentations · Residual Networks · Self-Supervised Learning

1 Introduction

According to [1], it is possible that environmental variations render the data inconceivable to models trained on data that do not capture these variations. Hence, the shifts in distribution of data in the inference stage of the fine-tuned model can lead to misclassifications of germinated seeds, leading to performance degradation of the model.

The environmental variations captured in the datasets, specified in Section 2, are pose and illumination of the germinated seeds. As this research is geared towards robust classification of germinated seed quality under the specified environmental variations, the dataset specifications allow us to posit a baseline model with a suitable data pipeline, then fine-tune the baseline architecture on Batch-1 with controlled variation in illumination and pose. It is noted that the existence of this dataset is a realisation of the environmental changes within data under real world conditions.

Past literature has agreed that high throughput of data collection is not necessarily feasible, and that data augmentation (DA) effectively mitigates this limitation for small datasets. DA also acts as a regularisation technique, improving model generalisations by introducing label-preserved augmented samples to the training pool, increasing the variation in the dataset [2]. It has also been mathematically proven that augmentations that form a dense distribution around the original image enhance the robustness of classifier models pertaining to vision tasks by tightening the generalisation gap and promoting the flatness of minima with respect to parameter space, but this assumes that the augmentations satisfy some conditions stated in the same paper. [3]

This report justifies the use of Convolutional Neural Network (CNN) architectures with some augmentations as a baseline. Failure modes accompanied by interpretability tools as a qualitative metric were identified within the baseline model. Subsequently, a self-supervised technique was employed to learn multi-view invariances. Metrics such as accuracy and F1 were computed to quantitatively compare the models.

2 Dataset

Three batches of germinated seed images were provided in this research. Batch-1 dataset consists of seeds derived from segmenting seeds with no pose variation and under a controlled LightBox environment. The specific details of the apparatus used in the experimental setup are included in the appendices. Batch-2 dataset consists of seeds under a normal room lighting condition while controlling for pose variation, while Batch-3 dataset consists of seeds

with varying poses under a controlled LightBox environment. Thus, Batch-2 and 3 are characterised primarily by the illumination and pose variation in seeds, respectively.

Each seed is labelled as a bad (0) or good (1) seed. These labels were derived by manual judgment of the seed quality, by observing the plumule and radicle component of the germinated seeds. A good seed is characterised by a more developed plumule and radicle. Hence, the datasets provided make this an ideal experimental setup to investigate the variations in isolation to hypothesise and evaluate methods for binary classification of germinated seeds under variations in illumination and pose.

3 Baseline evaluation

As the advantages of applying DA have been noted, a set of augmentations detailed in the appendices was applied to diversify training samples by introducing variations that may be present in Batch-2 and Batch-3. Nevertheless, performance degradation of the model is still expected on the downstream classification task as most samples from Batch-2 and Batch-3 remain OOD relatively to Batch-1, therefore exhibiting persistent domain shifts that is merely approximated by DA.

The authors of [4] question the relationship between the effectiveness of network learning to the number of layers in the network architecture. They have reported that stacking more layers in “plain” networks, such as VGG16 and variants, lead to a degradation in accuracy when modelling the system that was not due to overfitting, whereby training accuracy became saturated. Experimental results from the authors’ proposed architecture, residual nets (ResNet) on ImageNet showed that ResNet successfully overcame this obstacle — by inducing skip connections to solve the vanishing gradient problem in weight updates. ResNet50, compared to its counterparts varying in layer count (ResNet18, ResNet101, ResNet152) was preferred to strike a balance between architecture complexity and robustness to overfitting, especially with the size of the dataset on hand.

EigenCAM and GradCAM analysis were used as qualitative assessments of representations at each scale. EigenCAM produces a heatmaps corresponding to the first principle component (PC1) of the activations produced at that layer. Each loading of PC1 corresponds to a learned representation as a feature map; therefore, the final heatmap is interpreted as a linear combination of these feature maps weighted by their respective PC1 loadings [5]. High heat regions indicate strong, representative regions of the image to the activation of that layer. GradCAM on the other hand, produces visualisations of influential regions of the image to the class logits, making GradCAM analyses dependent on the classifier head used on the downstream task. Mathematically, salient regions imply that a perturbation in those regions correspond to a large change in the class logits relative to other regions of the image. Ideally, salient regions of EigenCAM show emphasis on meaningful foreground regions, such as the radicle and plumule, or indicate foreground and background differentiation, while GradCAM heatmaps show minimal salient spots on background artifacts, which are indicators of sensitivity to environmental shifts to the seed classification.

3.1 Preliminary analysis of ResNet50 on downstream task

In this section, pre-trained ResNet50 models were probed with a linear classifier head and a Multi-Layer Perceptron classifier head and underwent EigenCAM and GradCAM analysis to assess how each frozen layer captures the semantics of the seeds, and whether fine-tuning is needed for each layer, which, for ease of reference, will be denoted by L_i , $i = 1, 2, 3, 4$.

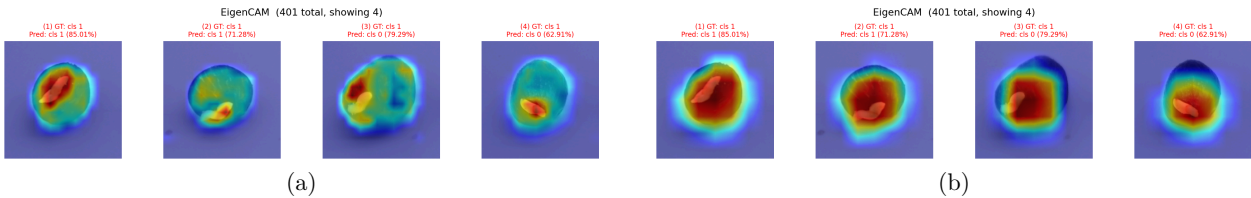


Fig. 1. EigenCAM heatmap samples of ResNet50 (a) L_3 (b) L_4 , sampled from Batch-1

In this work, the ResNet50 architecture utilised was pre-trained on ImageNet1K.V1, which consists of natural objects with edges and textures that transfers well to this domain. By replacing the final fully connected layer to produce two logits, ResNet50 has 23.5M parameters. Due to the limited size of this dataset, the analysis presented here is geared to justifying selectively tuning higher level layers that may not have been well adapted to the downstream task yet. Higher level layers that are not tuned to the dataset may be prone to poor learned representations, such as spurious correlations that minimise loss on the training set but have poor discriminative properties on illumination- and pose-varied images.

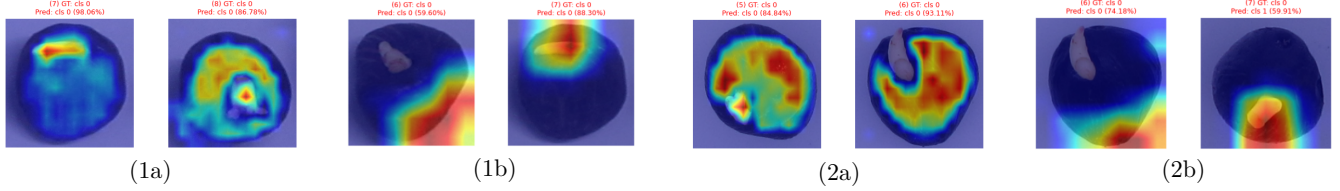


Fig. 2. EigenCAM heatmaps of (1) Batch-2 and (2) Batch-3 seeds from ResNet50 L_3 and L_4 : (a) L_3 and (b) L_4

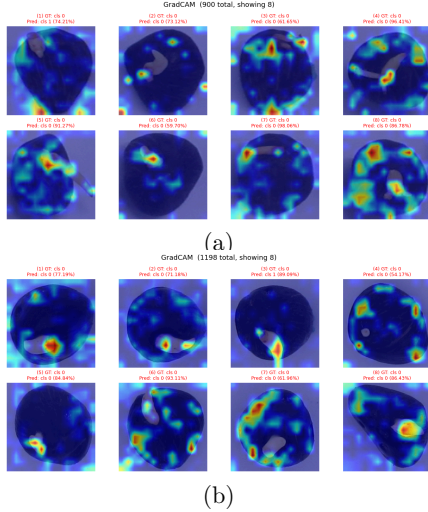


Fig. 3. GradCAM results from L_3 of ResNet50 + MLP head on (a) Batch-2 (b) Batch-3

The high-level feature maps extracted in L_4 were not consistently aligned to the downstream task, since there was a poor emphasis on the radicle and plumule regions, which are more specific, and higher-level semantics of the seed. While it is not as obvious in Batch-1 samples, EigenCAM analysis of ResNet50 reveal that there is a poorer focus on the radicle and plumule regions of the pose-varied seeds. Thus, in this work, L_4 is identified as a layer requiring fine-tuning, due to poor and inconsistent high-level semantic representation of the seeds.

In the subsequent layer, L_3 , the feature maps that dominate the PC1 representation of the activations of L_3 consistently overlap over the seed itself, which is evident from the EigenCAM samples from Batch-2 and Batch-3 in Fig. 2, which indicates that the model with a frozen backbone up to L_3 is capable of performing spatial localisation of the seed within the image itself. Some observed samples of EigenCAM were also able to localise the radicle and plumule region of the seed (see Fig. 2 (1)(a), left). However, for both classifier heads, the heat signatures of GradCAM appear scattered and poorly focused on the high-level semantics of the seed, which are the plumule and radicle, thus calling for further fine-tuning. As GradCAM heatmaps appear similar for both heads, only the heatmaps for the MLP head were shown.

3.2 Fine-tuned models and failure case analyses

Table 1. Table of Results on Batch-1

Model	<i>Acc</i>	<i>F1</i>	<i>AUC</i>
(1)	0.9726	0.9728	0.9982
(2)	0.9576	0.9592	0.9974
(3)	0.8479	0.8338	0.9319
(4)	0.8579	0.8593	0.9395

With the evidence presented, ResNet50 was fine tuned at layers 3 and 4, and a MLP classifier head was evaluated alongside a linear head on the downstream task. The tables above detail the results for (1) Fine-tuned ResNet50 + Linear head, (2) ResNet50 + 2-Layer MLP head, and (3), (4) are the respective fixed feature extractors. The MLP head is specified by $\text{MLP}(2048, 256, 2)^1$. Batch-normalisation was used as a regularisation technique in the MLP head, with a ReLU^2 activation function. Appendices detail the model training setup.

A few things can be noted from the results: overall, the linear head and MLP head performed similarly in terms of accuracy on all batches. However, the linear head always had a much higher

precision than the MLP head, while the MLP head always had a better recall. In all batches, the MLP generally had a better $F1^3$ score. (1) is taken as the baseline model.

Table 2. Results from Batch-2 and Batch-3

Model	Batch 2			Batch 3		
	<i>Acc</i>	<i>F1</i>	<i>AUC</i>	<i>Acc</i>	<i>F1</i>	<i>AUC</i>
(1)	0.8511	0.8378	0.9454	0.7838	0.7894	0.8677
(2)	0.8589	0.8630	0.9359	0.7718	0.7963	0.8479
(3)	0.7478	0.7577	0.8145	0.6167	0.6213	0.6532
(4)	0.7111	0.7275	0.7832	0.6054	0.6091	0.6627

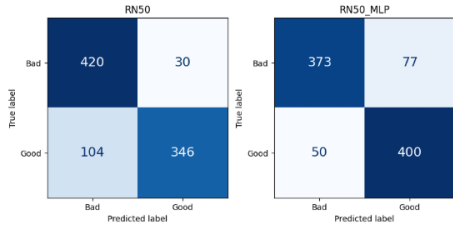


Fig. 4. Confusion matrices of baseline on Batch-2

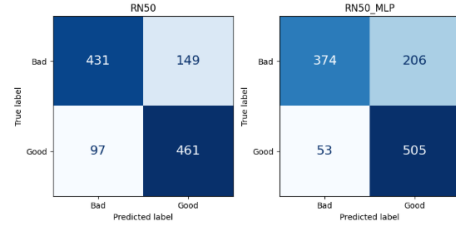


Fig. 5. Confusion matrices of baseline on Batch-3

¹ $\text{MLP}(a, h_1, h_2, \dots, h_n, b)$ denotes an MLP head with a input dimension, h_i hidden layer sizes, and b output dimension

²Rectified Linear Unit

³ $F1 = \frac{2}{\frac{1}{P} + \frac{1}{R}}$, where P is precision and R is recall.

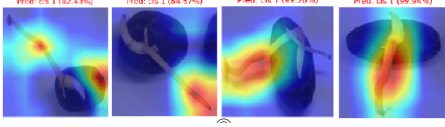


Fig. 6. GradCAM of representative false positive cases from Batch-2 (L_4)

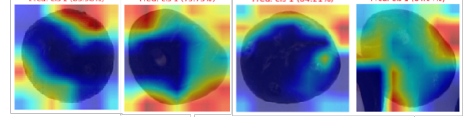


Fig. 7. GradCAM of representative false positive cases from Batch-3 (L_4)

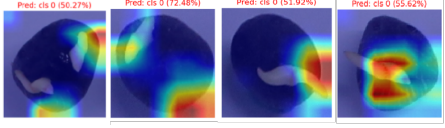


Fig. 8. GradCAM of representative false negative cases from Batch-2 (L_4)

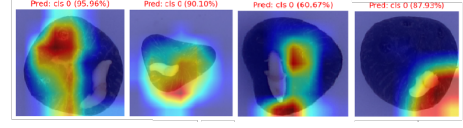


Fig. 9. GradCAM of representative false negative cases from Batch-3 (L_4)

Significant performance degradation of the baseline can still be observed on the downstream task when performed on Batch-2 and Batch-3, relative to Batch-1. In Batch-2, some false positive samples fell into with cases with long radicles protruding from the seed. In Batch-3, failure cases arose in seed images where no radicle or plumule are not visible at all on the seed, resulting in inconsistent salient maps that occasionally included hot regions in background regions, possibly mistaking the white background for a radicle. The GradCAM maps presented in Fig. 6 show that the salient regions did capture the radicle, but also captured the background, whereas those in Fig. 7 showed salient regions in the background, possibly mistaking the bright background behind pose-varied seeds as the radicle/plumule. This qualitative assessment shows that in many samples, the logits produced by the baseline model was still sensitive to background perturbations.

In both cases, many false negative samples had obvious plumules and radicles that were actually good seeds. High-level feature maps output at L_4 on Batch-2 images were not consistently salient at the radicle region, and saliency was present in the background of Batch-3 images.

K -means clustering with 2 clusters was also performed on the embeddings extracted from ResNet50 output fine tuned on Batch-1 with the linear head. Then, the adjusted random index (ARI) was computed for each batch, to measure similarities between clusters and ground-truth classes. The scores were $ARI_1 = 0.8930$, $ARI_2 = 0.4679$, and $ARI_3 = 0.3157$, for respective batches, which align with the current results. As clusters agree well with ground truth classes, a linear head is sufficient to perform well in classifying seeds with no illumination or pose invariance. On the subsequent batches with environmental shift, clusters on each class do not agree with the ground-truth classes well.

4 Hypothesis formulation

Self-supervised learning (SSL) follows a two-stage learning process by first minimising losses between multi-views simulated by augmentations, and then transferring the learned embeddings to a downstream task. The survey [6] described DAs as pretext tasks, and different categories of these pretext tasks are suitable for different scenarios. This aligns with the current work so far, as DAs must be carefully applied to avoid impossible samples when training the baseline model; random crops may not be suitable due to removing vital semantics of the seed that determine the seed class. On the other hand, colour transformations such as colour jitter, and geometric augmentations such as rotation are possibly useful to simulate illumination and pose-varying samples, respectively.

These constraints are less restrictive in SSL, since the training objective in SSL is class-agnostic and solely aimed to learn invariances between multiple views. Though, it is still wise to apply DA conservatively so as to not generate impossible views of the seed, which serve only as noisy samples for SSL [7]. Based on the ARI similarity metrics from the baseline model, the clusters on sample embeddings from Batch-2 and 3 agree poorly to the ground truth classes, and similarities saw significant degradation relative to Batch-1.

This work hypothesises that, by introducing a larger pool of augmentations, the SSL paradigm proposed here serves as consistency regularisation by training the encoder on invariant representations across plausible distribution shifts encountered augmentations, which is absent in supervised learning. Consequently, the embeddings yielded by the trained encoder are more consistent across batches, leading to a more robust model relative to the baseline. This further motivates the adoption of an SSL paradigm with a dual network architecture to obtain an encoder that learns invariances from paired multi-view images. The pre-trained encoder may be subsequently fine-tuned on the downstream task for classification to evaluate whether this hypothesis holds. Additionally, this hypothesis could be falsified if cluster to ground-truth class alignment based on ARI metric was poor in general or still experienced similar degradation across Batch-2 and 3 compared to the baseline. These observations should also align with accuracy degradation across the batches.

5 Self-supervised Learning with BYOL

Contrastive methods rely on large batch sizes to provide more negative pairs per batch [8], memory banks or negative mining strategies to retrieve hard negative samples to produce sufficiently large gradient magnitudes for a low computational cost. This study bypasses the need of negative samples by applying an implementation⁴ of one *Bootstrap Your Own Latent* (BYOL) that incorporates an asymmetric architecture minimising a mean squared error between positive pairs' embeddings produced by two networks, an online, and a target network. [9] These networks are denoted by f_θ and f_ξ , with network parameters θ and ξ , respectively.

$$\begin{aligned}\mathcal{L}_{\theta,\xi}(v, v') &\triangleq \|\hat{q}_\theta(z_\theta) - z'_\xi\|_2^2 \quad 5 \\ &= 2 - 2 \cdot \hat{q}_\theta(z_\theta)^\top \hat{z}'_\xi\end{aligned}\tag{5.0.1}$$

$$\begin{aligned}\theta &\leftarrow \text{SGD}(\theta, \nabla_\theta \mathcal{L}_{\theta,\xi}, \eta) \\ \xi &\leftarrow \tau \xi + (1 - \tau) \theta\end{aligned}\tag{5.0.2}$$

Note that, in both networks, θ and ξ encompass the encoder and projector weights, while θ also encompass the predictor model. Without negative pairs, the original work in 2020 has shown that BYOL produced on SOTA results on various benchmarks, without learning collapsed representations.

While the dual encoder architecture effectively doubles the GMACs⁶ and parameter count during the training stage compared to supervised learning as in the baseline model, the dynamics of BYOL as presented in (5.0.2) ($\tau = 0.996$ in this work) ensures that optimisation is only explicitly done on θ , not ξ , hence, trainable params are not doubled. In the same equation, the teacher network with parameters ξ are updated by a slow-moving average, producing targets for the online network. Baseline model requires 4.09 GMACs, per input, while the BYOL architecture has 8.18 GMACs. Both the online and target network use a MLP to project the 2048-dimensional embedding to 128 dimensions. Additionally, the online network introduces an additional predictor head, denoted by q_θ , to predict the high-dimensional embedding output of the teacher network. The projector head g_θ is MLP(2048, 512, 128), and q_θ is MLP(128, 512, 128). Hence, the student encoder is trained to match view v to view v' , and vice versa, by minimising the symmetric loss $\tilde{\mathcal{L}}_{\theta,\xi}(v, v') = \mathcal{L}_{\theta,\xi}(v, v') + \mathcal{L}_{\theta,\xi}(v', v)$. Simplification of the loss function also shows that minimising the loss $\tilde{\mathcal{L}}_{\theta,\xi}(v, v')$ is effectively maximising the cosine similarity between embeddings from views v and v' . $\mathcal{L}_{\theta,\xi} = 0 \implies$ exact embedding alignment, while $\mathcal{L}_{\theta,\xi} = 4 \implies$ orthogonal embeddings.

The stochastic pipeline of BYOL introduces more aggressive augmentations such as a GaussianBlur and RandomResizedCrop, to increase diversity of training pairs. These aggressive augmentations are not suitable when training an encoder via supervised learning, as these augmentations may not preserve the semantic features required to predict the label. Directly applied to supervised learning on the downstream task, these augmentations may harm model performance.

Instead of a LARS optimiser as utilised by the original work in their benchmarks, the effectiveness of LARS lies in using large batch sizes (4096) on large datasets. As this work only uses a manageable batch size of only 32, a SGD optimiser was used. Further details of the BYOL architecture and training setup are included in the

⁴<https://github.com/deepmancer/byol-pytorch>

⁵ $\hat{q}_\theta(z_\theta) = q_\theta(z_\theta) / \|q_\theta(z_\theta)\|_2$

⁶GMACs: Giga Multiply-Accumulate Computations per Second

appendices. The loss curve in Fig. 10 was obtained after training the encoder over 100 epochs, achieving the loss of $\tilde{\mathcal{L}}_{\theta,\xi} = 0.1522$.

A linear probe on the BYOL online encoder trained over 150 epochs yielded subpar accuracy on Batch-1 and Batch-2 relative to the baseline with a linear head, though performance was more consistent across batches, suggested by the smaller decrease in accuracy across batches compared to the linear probe on the ResNet50 fixed feature extractor. ARI metric tabulated in Table 3 show that embeddings behaved more similarly across batches than the linear probe baseline, in the weak sense that clusters aligned similarly with ground-truth classes across batches, unlike the linear probe baseline, whereby ARI metric degraded even more significantly.

Nonetheless, the results of the linear probe on the online encoder was still poor, and required fine-tuning on the downstream task, though this may cause catastrophic forgetting to some capacity. To reduce this effect, fine-tuning was done only on 50 epochs with similar training setup as baseline detailed in appendices, to reduce the number of weight updates of the model. It can be seen that this barely interrupts training as accuracy history plots in Fig. 11 show that the validation accuracy has already converged at this epoch.

The results tabulated in Table 3 show that the fine-tuned online encoder consistently outperformed the baseline across all batches. The combined training history shows that the online network reached a higher training ceiling earlier, with validation accuracy converging more smoothly and a smaller train-validation gap throughout, suggesting better generalisation and reduced overfitting on Batch-1. K -means clustering with $K = 2$ was further applied to the fine-tuned encoder, with ARI scores for both the baseline and BYOL online encoder also tabulated in Table 3. Cluster of online encoder embeddings to ground-truth alignment was more consistent across batches but still showed some degradation.

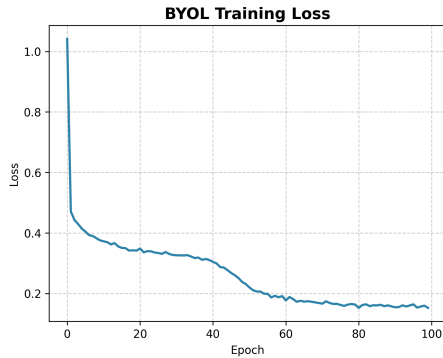


Fig. 10. Training loss curve from BYOL

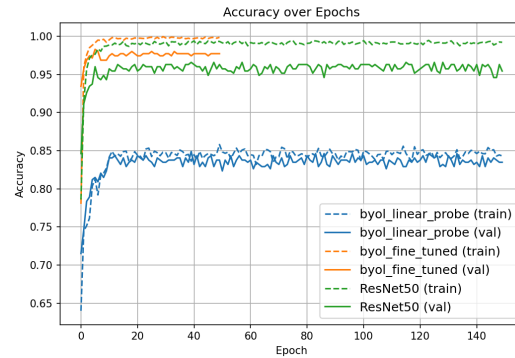


Fig. 11. Accuracy history plots

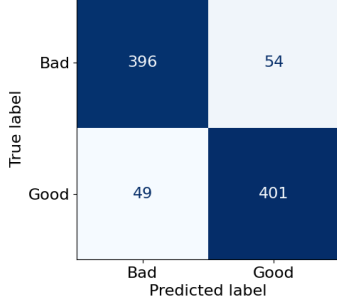


Fig. 12. Confusion Matrix of online encoder with linear head on batch-2

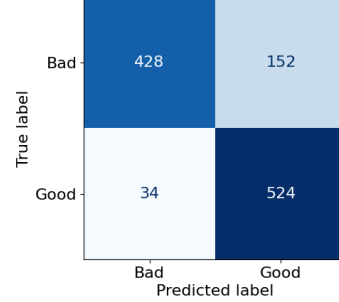


Fig. 13. Confusion Matrix of online encoder with linear head on batch-3

Table 3. Table of metrics across all batches. Bracketed values indicate drop in accuracy from Batch-1

	Batch-1				Batch-2				Batch-3			
	Acc	F1	AUC	ARI	Acc	F1	AUC	ARI	Acc	F1	AUC	ARI
<i>Linear Probe</i>												
Base-line	0.8479	0.8338	0.9319	0.1930	0.7478 (0.1001)	0.7577	0.8145	0.0130	0.6160 (0.2319)	0.6183	0.6515	-0.0010
BYOL	0.7880	0.7432	0.9319	0.0990	0.7111 (0.0769)	0.7561	0.7955	0.0290	0.6336 (0.1544)	0.6475	0.6733	0.0200
<i>Fine-tuned</i>												
Base-line	0.9726	0.9728	0.9982	0.8930	0.8511 (0.1215)	0.8378	0.9454	0.4679	0.7838 (0.1888)	0.7894	0.8677	0.3157
BYOL	0.9850	0.9852	0.9995	0.9216	0.8856 (0.0994)	0.8862	0.9501	0.5406	0.8366 (0.1484)	0.8493	0.9197	0.4133

6 Conclusion, limitations and future work

In conclusion, this work presents a fine-tuned ResNet50 with linear head, trained on label-preserving augmentations as a baseline model. Saliency maps revealed inconsistencies in regions of emphasis across environmentally shifted batches, suggesting that the baseline model is sensitive to spurious background features rather than seed semantics. K -means clustering ($K = 2$) results on high-dimensional embeddings from the baseline encoder were used to compute Adjusted Random Index (ARI), poor similarity between clusters and ground-truth class on Batch-2 and 3 relative to Batch-1, indicating that the embeddings are not yet consistent in environmentally shifted seed images. A limitation of training an encoder model in a supervised manner is sensitivity to data augmentation types applied in the pipeline, since aggressive augmentations may erase semantics of the seed explanatory to the seed class.

The hypothesis formulated led to applying *Bootstrap Your Own Latent* (BYOL) as a self-supervised learning (SSL) technique for consistency regularisation of embeddings across batches. The encoder model was trained on the pretext task of learning invariances under data augmentations. The encoder was then assessed under a linear probe and also fine-tuned on the downstream task, which showed poor accuracy and ARI metric relative to the linear probe baseline, but the said metrics were more consistent throughout batches for the former. This prompted further fine-tuning on the downstream task, which was acknowledged to potentially cause some catastrophic forgetting

of learned invariances from SSL. However, results showed that the fine-tuned online encoder yielded superior performance and robustness to the baseline model across all batches. By computing ARI in a similar manner, K -means clusters had a better agreement to the ground-truth classes than the baseline — ARI metrics were more consistent but still saw degradation across batches. This highlights a fundamental limitation of the final model — although a more performant final model could be obtained by applying SSL as a preliminary step, a trade-off between learned invariances and downstream performance must be made.

This work showed some success in applying BYOL for robust modelling of seed quality classes. Application of BYOL in this study noted that augmentations were still conservative to ensure plausible views of the seed to learn invariances. Future work could apply MSR by the authors of [7] in which both weaker and more aggressive augmentations under a modified framework with an asymmetric architecture similar to BYOL, which was more robust to aggressive augmentations.

7 Appendices

7.1 Appendix A: Dataset

Batch-1 consists of 110 images of good quality seeds and 105 images of bad quality seeds. Each image contains around 10-11 individual seeds with a clear gap between seeds. Segmented seeds readily prepared for analysis were made use for model training. Listed below are the related apparatus:

1. Compact digital camera Samsung NX2000 camera for capturing images, set to autofocus
2. Tripod to fix digital camera 40 cm away from the light box
3. Sanoto MK40 Photo Light Box was used as a tool to emit light under light box condition.

Batch-2 and 3 images were collected under the below listed conditions:

1. Batch-2 : Phillips Lifemax TLD 36W/54-765 fluorescent bulb.
2. Batch-3 : Sanoto MK40 Photo Light Box

with the listed apparatus:

1. Compact digital camera Canon IXUS 285 HS, set to autofocus function, to capture images. The camera was set 26 cm above a table top where the seeds were positioned.

Each image from Batch-2 and 3 consist of multiple good/bad/mixed seeds in a line or randomly spread on a white background. Evaluation was done on 900 seeds from Batch-2 and 1138 seeds from Batch-3.

Annotations of Batch-2 and 3 seeds were also included as .xml files, which consisted of ‘Good Seed’ or ‘Bad Seed’ labels with each seed’s ‘Rectangular Bounding Box’.

7.2 Appendix B: Augmentations and training setup of baseline

- Aside from necessary augmentations such as resize to (224, 224) for and normalisation for ResNet50 input, augmentation pipeline for supervised learning of baseline models include RandomHorizontalFlip($p = 0.5$), RandomRotation θ randomly sampled from $U[0^\circ, 30^\circ]$, ColorJitter with brightness and contrast sampled from $U[0.6, 1.4]$.
- Models (1) - (4) in Section 3.2 were trained under similar setup with stochastic gradient descent (SGD) optimiser, with exponential decay learning rate scheduler every 10 epochs, $\gamma = 0.1$, over 150 epochs.
- MLP head used batch normalisation as a regularisation method. ReLU was the activation function.

7.3 Appendix C: BYOL

BYOL Architecture

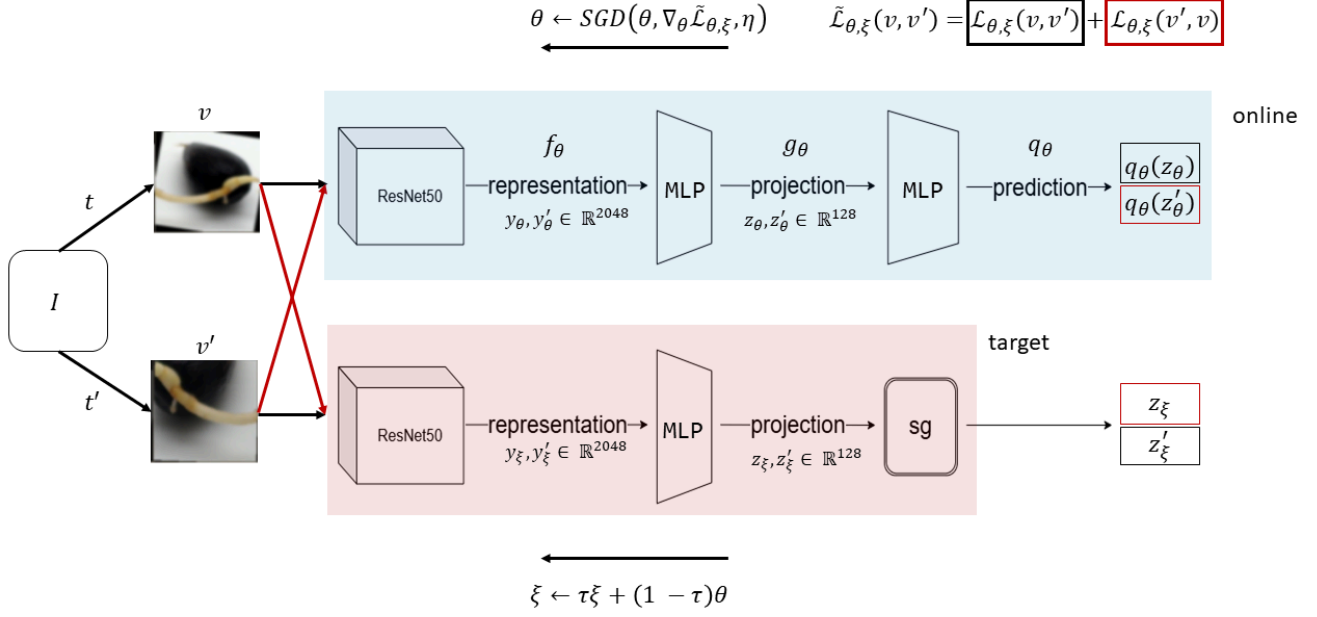


Fig. 14. BYOL architecture. It is an asymmetrical dual encoder architecture with an online network (f_θ) and target network (f_ξ). [9]

The symmetrical loss $\tilde{\mathcal{L}}_{\theta,\xi}$ is computed by cross-view of augmented samples of the original image I then backpropagated only in the student network to update weights. Thus, **sg** is imposed on the teacher network and its weights are instead updated by an exponential moving average.

BYOL augmentations**Table 4.** BYOL augmentation pipelines for View 1 and View 2.

Transformation	View 1, v	View 2, v'
Random Resized Crop (scale)	80% chance, (0.2, 0.9)	20% chance, (0.6, 0.9)
Resize	224×224	224×224
Random Horizontal Flip	$p = 0.5$	$p = 0.5$
Color Jitter (brightness, contrast)	20% chance, $U(0.6, 1.4)$	80% chance, $U(0.6, 1.4)$
Random Rotation	sampled from $U(-30^\circ, +30^\circ)$	sampled from $U(-30^\circ, +30^\circ)$
Gaussian Blur ($\sigma \in [0.1, 2.0]$)	$p = 1.0$	$p = 0.3$

BYOL training setup

- A SGD optimiser was used with a learning rate of 0.002 and momentum of 0.9
- Learning rate scheduler used was a cosine learning rate scheduler with $\eta_{\min} = 0$ over 100 epochs.

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